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MATH2056 - LINEAR ALGEBRA

MAKEUP FINAL EXAM =)

"On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues."

QUESTIONS =)

① Consider the matrix $A = \begin{bmatrix} 1 & \alpha & -\gamma & \alpha \\ 3\alpha & 5 & -9\gamma & 1 \\ 2 & \beta & 3 & \alpha^2 \\ 1 & 14 & -6 & 2 \end{bmatrix}$. Some of the elimination matrices are given below:

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & -1 & 1 & 0 \\ 0 & k & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ -8 & 1 & 0 & 0 \\ m & 0 & 1 & 0 \\ -1 & 0 & 0 & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & -1/2 & 1 \end{bmatrix}$$

a) (5p.) Find the unknown entries α, γ, β, k and m = ?

b) (5p.) Find the eliminated upper triangle matrix U = ?

c) (10p.) Determine the remaining elimination matrices!.

② A geometry in 'k' dimensional space is given by $x = 5y$ expression, where 'x' and 'y' represents the axis. The intersection with another geometry gives a volume.

a) (10p.) Give the minimum dimension of the space with your valuable comments.

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b) (10p.) Propose an expression for the unknown geometry to have a volume in intersection.

c) (15p.) Proposed a geometry $x+4y+z=6$. Find all vectors in the intersection. Do these vectors form a vector space? The intersection has how many dimensions inside how many dimensional space?

(3) Suppose u is a unit vector in R^n. This problem is about the n by n symmetric matrix $H = I - 2uu^T$:

a) (5p.) Show directly that $H^2 = I$

b) (5p.) Show that H is an orthonormal matrix.

c) (5p.) One eigen vector of H is u itself. Find the corresponding eigenvalue.

d) (5p.) Is H diagonalizable? Why or why not?

(4) Two tanks are connected, first tank has $v(0)=10$ liters liquid and the second tank has $w(0)=40$ liters of water. The liquid diffusion between the tanks are proportional to the difference $v-w$.

a) (10p.) Which one of the following differential equation set expresses the process? (v_k: the liquid diffused to the second tank, w_k: the water diffused to the second tank.)

$$\begin{bmatrix} v_{k+1} \\ w_{k+1} \end{bmatrix} = \begin{bmatrix} -2 & 1 \\ 1 & -2 \end{bmatrix} \begin{bmatrix} v_k \\ w_k \end{bmatrix} \qquad \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix} \qquad \frac{d}{dt}\begin{bmatrix} v \\ w \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} v \\ w \end{bmatrix}$$

b) (10p.) Determine percentages of the mixture in tank1 at time $t=1$ sec.

c) (10p.) Find how much liters of liquids are exist in each tank in steady state?

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SOLUTIONS=)

(4) a) Birinci tank daha az sıvıyla başladığı için sıvı aktarımı birinci tanka doğru olur. Birinci tankta sıvı artışı, ikinci tankta sıvı azalışı gözlenir.

$$\Rightarrow V_{k+1} = V_k + (V_k - W_k)t \qquad W_{k+1} = W_k - (V_k - W_k)t$$

$$\frac{dV_{k+1}}{dt} = \frac{d\,(V_k + (V_k - W_k)t)}{dt} \qquad \frac{dW_{k+1}}{dt} = \frac{d\,(W_k - (V_k - W_k)t)}{dt}$$

$$\frac{dV}{dt} = 0 + V_k - W_k \qquad \frac{dW}{dt} = 0 - V_k + W_k$$

$$\Rightarrow \begin{bmatrix} V_{k+1} \\ W_{k+1} \end{bmatrix} = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \begin{bmatrix} V_k \\ W_k \end{bmatrix} \quad \& \quad \frac{d}{dt}\begin{bmatrix} V \\ W \end{bmatrix} = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} V \\ W \end{bmatrix}$$

Bu değerlere en yakın seçenek: $\frac{d}{dt}\begin{bmatrix} V \\ W \end{bmatrix} = \begin{bmatrix} 2 & -2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} V \\ W \end{bmatrix}$ //

b) Tank 1 → V_k ↓ W_k ↑

$$V_t = 10 + (40-10).k \qquad V_t \text{ in tank1} = 10 + (40-10)k.40$$

$$W_t = 40 - (40-10).k \qquad W_t \text{ in tank1} = (40-10).k.40$$

$$\frac{10 - (40-10).k.10}{(40-10).k.40} = \frac{V_1}{W_1} = \frac{10 - 300k}{1200k}$$ //